Predicting Good Solar Hours for Energy Planning from Weather and Location Data

Milestone 2: Evaluation plan

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k nearest neighbours with standardisation and k = 11 reaches 73.8 percent accuracy on sites it has not seen, against a baseline of 44.8 percent, but still calls 31.3 percent of the weak hours good. Every number in this report is an output of notebooks/02_evaluation.ipynb. The test part is cut off and stays closed: no score is computed on it.

Part A. A first model

Field 1. Model

We use k nearest neighbours. The model contains no rule about sunshine. It stores the known hours with their labels, and for a new hour it finds the k known hours that are most similar and gives the majority of their labels as the answer. Similar means close: for every one of the 11 inputs we take the difference between the two hours, square it, add the squares and take the square root. This is the straight-line distance.

$$d(Q, A) = \sqrt{((q_1 - a_1)^2 + (q_2 - a_2)^2 + \ldots + (q_{11} - a_{11})^2)}$$

An example from our data: Q is a weak hour at Peterson, in June at 13:00, with clouds at 1.3 km. In round 3 of the cross validation, Peterson is hidden, so the model has never seen Q's label. Its three closest known hours, A, B and C, are all from USAFA, a site in the same city.

Table 1. Q and its three nearest known hours (raw inputs)

Hour	Site	Label	Distance to Q
Q	Peterson	weak (hidden)	0
A	USAFA	good	69.2
B	USAFA	weak	69.2
C	USAFA	good	69.3

With k = 1, only A votes, so Q is called good, which is wrong. With k = 3, A and C outvote B, so the answer is again good. On two inputs only, humidity and temperature, the distance from Q to A is $\sqrt{(2.16^2 + 5.18^2)} = 5.6$, to B 2.9 and to C 7.6. On these two inputs, the weak hour B is the closest, and k = 1 would answer correctly. Why A wins on all 11 inputs is the subject of Field 5.

The model has three hyperparameters, which we choose and the model does not learn from the data: the number of neighbours k (n_neighbors), the distance measure (straight-line distance, the default in scikit-learn) and the weight of each neighbour's vote (weights="uniform", every neighbour counts equally). We start with k = 1 and choose k in Field 6.

Part B. Measuring honestly

Field 2. Split

We hold out about 20 percent of the data as the test part, a common compromise between a reliable final score and enough examples for training. Our users are factories at places the model has never seen. Rows of one site are related: they share the same coordinates and the same local climate. If a site appeared on both sides of the cut, the test would measure how well the model knows that site, not how well it works at a new place.

We therefore cut by site, with StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42) and the column Location as the group. It distributes the 12 sites into 5 groups without ever splitting a site, keeps the share of good hours in each group as close to the total as whole sites allow, and we take the first group as the test part. random_state=42 makes the same sites land in the test part on every run and every computer.

Table 2. Training and test part

Part	Sites	Rows	Share of data	Good hours
Training	10 sites	16,760	the rest	51.2%
Test	Grissom, Travis	4,233	20.2%	45.6%

The shares of good hours cannot match exactly, because a whole site must stay on one side: Grissom has only 29.3 percent good hours, Travis 54.4 percent. Both test sites cover all 12 months, so the test part is not limited to one season. Keeping each site whole also keeps all readings of the same site and day together. The test part stays closed until the final test at the end of the project.

Field 3. Baseline

The baseline is DummyClassifier(), a lazy model that ignores the weather and always answers the class that is more common in its training data. We measure it by cross validation on the training part, with the same five folds of whole sites as our model (Field 4). Its accuracy per fold is 40.1, 47.4, 42.0, 44.9 and 49.5 percent, with a mean of 44.8 percent and a spread of 3.4 points.

This is below 50 percent because the two classes are almost equal: hiding a group of sites pushes the remaining sites slightly toward the opposite class, so the lazy model learns the wrong majority in every round. For example, when JDMT and MNANG are hidden (together 59.9 percent good hours), the rest has only 49.6 percent good hours, so the lazy model answers "weak" for every hour and is right only 40.1 percent of the time.

Our second measure is the share of weak hours wrongly called good. Over the five rounds, the four outcomes are:

Table 3. Four outcomes of the baseline

Truth	Says weak	Says good
Weak hour (8,184)	3,251 correct	4,933 called good
Good hour (8,576)	4,265 missed	4,311 correct

4,933 of 8,184 weak hours, 60.3 percent, are wrongly called good. Every real model must beat 44.8 percent accuracy and stay well below 60.3 percent on the second measure.

Field 4. Cross validation

We use five folds inside the training part, made with the same tool as the split: StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42) with the site as the group. The folds are shuffled, never split a site, and are stratified as far as whole sites allow. The baseline in Field 3 uses exactly the same folds, so the two can be compared fold by fold.

Table 4. Accuracy per fold: baseline and k = 1 on raw inputs

Round	Hidden sites	Baseline	k = 1, raw inputs
1	JDMT, MNANG	40.1%	66.8%
2	Kahului, Malmstrom, Offutt	47.4%	68.6%
3	Peterson	42.0%	67.2%
4	Hill Weber, USAFA	44.9%	59.1%
5	Camp Murray, March AFB	49.5%	58.7%
Mean (spread)		**44.8% (3.4)**	**64.1% (4.3)**

k nearest neighbours with k = 1, on the raw inputs without any preparation, reaches a mean accuracy of 64.1 percent with a spread of 4.3 points. It beats the baseline in every fold, by 19.3 points on average, far more than the spread. Of the 8,184 weak hours, 2,733 (33.4 percent) are wrongly called good, about half the 60.3 percent of the baseline. Still, one in three weak hours would lead the factory to schedule work without enough sun, so the model is not yet good enough for real use.

Part C. Improving without cheating

Field 5. Preparation

A concrete mistake: Q from Field 1, a weak hour at Peterson held out in round 3, is called good by k = 1 on the raw inputs. Its nearest known hour A is a good USAFA hour at distance 69.2, while the weak USAFA hour B is only slightly farther. The cause is the units of the inputs. Table 5 shows what each input adds to the squared distance.

Table 5. Contribution of each input to the squared distance from Q

Input	Q to A (good)	Q to B (weak)
Altitude	4,624.0	4,624.0
Pressure	118.8	148.8
AmbientTemp	26.9	6.5
Wind.Speed	10.2	10.2
Humidity	4.7	1.6
Month	1.0	0.0
Hour	1.0	1.0

Altitude alone, a difference of 68 m, makes up 96.6 percent of the squared distance to A. Altitude is the same for every hour of a site, so it only decides which site the neighbours come from. Within USAFA, Pressure decides: B is closer to Q in temperature, humidity and month, but farther in pressure, so A wins. Temperature, humidity and cloud ceiling, the inputs that matter most for solar power, hardly count because their numbers are small.

The preparation is standardisation: every value is replaced by its distance from the column mean, measured in standard deviations of that column.

$$z = (\text{value} - \text{mean of the column}) \div \text{standard deviation of the column}$$

In round 3, the scaler learns a standard deviation of 713.0 m for Altitude and 12.7 °C for AmbientTemp. The altitude difference of 68 m becomes 0.10, the temperature difference of 5.2 °C becomes 0.41. After standardisation, Q's three nearest neighbours are weak USAFA hours at distances 0.44, 0.54 and 0.56, so Q is answered correctly.

The scaler learns its means and standard deviations from data, so it must be fitted inside each round. If it learned them from all training rows, the hidden sites of a round would shape the preparation before being asked about, which is a leak. We therefore put StandardScaler and the model into one Pipeline: in every round, both learn only from the four known folds.

Table 6. Accuracy per fold before and after standardisation (k = 1)

Round	Raw	Standardised	Difference (points)
1	66.8%	66.1%	−0.7
2	68.6%	71.0%	+2.4
3	67.2%	66.5%	−0.7
4	59.1%	74.4%	+15.3
5	58.7%	72.8%	+14.1
Mean (spread)	**64.1% (4.3)**	**70.2% (3.3)**	

The mean rises from 64.1 to 70.2 percent, a gain of 6.1 points, larger than the spread. The gain comes from rounds 4 and 5; in rounds 1 and 3 accuracy drops slightly, by less than the spread. Standardisation does not reduce the worse mistake: 2,717 of 8,184 weak hours (33.2 percent) are still called good, against 33.4 percent before. The gain comes from good hours that are now recognised: 2,159 missed instead of 3,411.

Field 6. Tuning

We run a grid search over the number of neighbours k, with the values 1, 3, 5, 7, 9, 11, 15, 21, 31 and 51 (odd, so that votes cannot tie). Each value uses the same pipeline (standardisation and k nearest neighbours) and the same five folds of whole sites, all inside the training part.

Table 7. Grid search over k

k	Accuracy	Spread (points)	Weak hours called good	Good hours recognised
1	70.2%	3.3	33.2%	74.8%
3	71.5%	3.2	32.4%	76.7%
5	72.9%	2.9	31.6%	78.5%

k	Accuracy	Spread (points)	Weak hours called good	Good hours recognised
7	73.5%	2.5	31.4%	79.3%
9	73.7%	2.4	31.3%	79.7%
11 (chosen)	**73.8%**	**2.3**	**31.3%**	**79.8%**
15	74.0%	2.2	31.4%	80.2%
21	74.0%	2.3	32.0%	80.8%
31	74.4%	2.2	32.1%	81.5%
51	74.5%	2.5	32.0%	81.7%

Accuracy rises from 70.2 percent at k = 1 to 73.7 percent at k = 9, more than the spread. From k = 7 to k = 51, all values lie within 1 point of each other (73.5 to 74.5 percent), which is smaller than the spread of about 2.3 points, so these differences are not convincing. The grid search picks k = 51 by accuracy, but only by such a small difference.

We choose k = 11. Its accuracy of 73.8 percent is within 0.7 points of the best, less than the spread, and it has the lowest share of weak hours called good, 31.3 percent (tied with k = 9), which Milestone 1 named as the worse mistake. This share hardly depends on k: it stays between 31.3 and 33.2 percent for every value. Choosing k reduces the missed good hours, but about one in three weak hours is still called good, and no value of k changes that.